# Q1 — Backend Answers

ShelfLife — IA2 Full Stack Web Development

## Assumptions

The application has a basic librarian-only authentication layer. Members are library records and do not log in to the frontend. The implementation focuses on the tasks explicitly required in the assessment.

## Q1(a) — Mongoose schemas

Three MongoDB/Mongoose models are used: Book, Member and BorrowRecord. Book stores title, author, ISBN, genre, totalCopies and availableCopies. Member stores name, email, membershipId and joinedDate. BorrowRecord references a Book and Member and stores issueDate, dueDate, returnDate and status.

Required fields and uniqueness/validation rules are applied where required, including unique ISBN and email, and references are used between borrowing records and their related book/member.

## Q1(b) — REST API

POST /api/books — adds a new book.

GET /api/books — lists books with pagination and genre filtering.

POST /api/members — registers a new member.

POST /api/borrow — issues a book after checking that availableCopies is greater than zero, decreases availableCopies, and creates a BorrowRecord.

POST /api/return/:borrowId — returns a book, records returnDate, increases availableCopies and updates the borrow status.

GET /api/members/:id/history — returns the member's complete borrowing history.

## Q1(c) — Middleware

Request logging middleware records incoming HTTP method and URL.

Input validation is implemented using simple manual checks rather than Joi/Zod. Required request fields are checked before controller logic executes.

Centralized error-handling middleware receives errors from the application and returns a consistent JSON error response.

## Q1(d) — Authentication

POST /api/auth/login checks the librarian credentials and issues a JWT. Protected librarian operations require the JWT in the Authorization: Bearer header.

## Q1(e) — Race condition

To prevent two librarians from issuing the last copy at the same time, the availability check and decrement should be one atomic database operation.

MongoDB can update the book only when availableCopies > 0 and decrement availableCopies using $inc.

If two requests arrive for the final copy, only one update can satisfy the condition; the other request fails.

This prevents availableCopies from becoming negative without requiring a distributed lock or a queue for every issue request.

## Implementation scope

Only the tasks required by the assessment have been implemented. Extra library-management features were intentionally not added.